

Classification Algorithm v2

Transparent eight-category, multi-label baseline | Prepared by the eight-member CS-44 project team

The CS-44 project team completed the taxonomy configuration, scoring design, evidence review, quality assurance, and report production. Categories are not mutually exclusive. Simulation and Case-Based Learning are separate. Week 4 remains an INFS6600-only rerun.

1. Implemented Week 4 changes

- Split Simulation from Case-Based Learning.
- Allow a unit and an evidence item to belong to more than one category.
- Expect INFS6600 in Work-Integrated and Applied, Project- and Problem-Based, and Case-Based Learning.
- Add category-level total scoring.
- Configure the remaining supplied taxonomy categories without analysing additional units.

Supporting pilot refinements

- Count distinct evidence items rather than raw phrase occurrences.
- Retain overlapping industry/case evidence instead of forcing a single label.

2. Evidence unit and counting policy

One distinct outline item (overview paragraph, learning outcome, assessment row, or weekly-schedule row) evaluated once per category.

Each outline item is evaluated once per category. A rule contributes its weight once even when several alternative phrases are present. Deduplication is at item-category level, so the same item can count once in several categories.

3. Taxonomy v2

Category	Definition	Overlap / review rule
Work-Integrated and Applied Learning	An educational approach that explicitly merges theory with real-world practice and embeds authentic industry, workplace, or community-relevant work and tasks into a unit of study.	May overlap with Project- and Problem-Based Learning, Case-Based Learning, Community Learning, and Entrepreneurial Learning. Check that professional or partner language describes an authentic learning task rather than a generic context statement.
Simulation	An instructional method that recreates or mimics real-world scenarios, environments, or processes so learners can practise skills and make decisions safely.	Separated from Case-Based Learning in the Week 4 taxonomy. A real-world scenario alone is not a simulation unless the activity recreates, mimics, or role-plays a setting or process. Do not inherit Simulation from a case-study label or from Case-Based Learning evidence.

Category	Definition	Overlap / review rule
Case-Based Learning	An active teaching method in which students use real-world scenarios and case studies to build critical-thinking and problem-solving skills.	May overlap with Work-Integrated and Applied Learning and Project- and Problem-Based Learning. Industry context does not exclude a case-based classification. An administrative assessment type such as 'Case studies' is retained as a weak review signal and is not sufficient by itself for a positive classification.
Project- and Problem-Based Learning	Learning focused on real-world problems and challenges using problem-solving, decision-making, investigative, and self-directed learning skills.	May overlap with Work-Integrated and Applied Learning when projects involve clients or industry partners, and with Case-Based Learning when cases are used inside a project. Do not classify from the generic word 'project' alone; require a problem, challenge, immersion, investigation, or explicit project/problem-based method.
Community Learning	Learning undertaken together with or for communities, including service-learning, field studies, and professional communities of practice.	May overlap with Work-Integrated and Applied Learning when the authentic partner is a community organisation. Require an explicit community, service-learning, field-study, or community-of-practice activity.
Entrepreneurial Learning	Learning that develops the knowledge, skills, and mindset to recognise opportunities, turn ideas into action, and manage new ventures under uncertainty.	May overlap with Project- and Problem-Based Learning and Simulation when learners develop or simulate a venture. Generic ideation, pitching, or prototyping is a review signal unless explicitly connected to entrepreneurship, a startup, a product-development project, or a venture.
Technology-Mediated Learning	An instructional environment in which digital devices, networks, or specialised technologies mediate access to content and interaction during learning.	May overlap with Simulation, Hybrid Learning, or Entrepreneurial Learning. Merely studying digital technology is not evidence that learning is technology-mediated. Require the technology to be part of the learning method, not only the subject matter.
Hybrid Learning	An education model that mixes face-to-face classroom teaching with online digital instruction.	May overlap with Technology-Mediated Learning, but requires an explicit combination of physical and online modes. Online materials alone are insufficient; require an explicit hybrid, blended, flipped, hyflex, or mixed-mode design.

4. WIL scoring weights

Rule group	v2 weight	Status
industry or business partner	4.0	existing_high_weight
actual organisation or professional context	4.0	existing_high_weight
authentic practice	4.0	team_calibrated_4
professional presentation	3.5	unchanged
theory-practice integration	2.0	team_calibrated_2
practical teamwork	2.0	team_calibrated_2
career readiness	2.0	provisional_team_calibrated_2
project immersion	2.0	unchanged

5. Decision thresholds and confidence

Positive ≥ 3.0 ; high confidence ≥ 5.0 . Status: provisional_pending_validation. These values remain configurable and are not represented as empirically validated cut-offs.

6. Scoring and aggregation

- Normalise lowercase, whitespace, apostrophes, and dash variants.
- Sum weights for matched rule groups within one item-category decision.
- Send an item to review when it is below the positive threshold or contains only review-required rules.
- Classified score total = sum of positive item scores for the unit-category pair.
- Review score total is shown separately and never silently added to the positive score.

7. Case-Based versus Simulation

Open-ended business scenarios support Case-Based Learning. A Simulation result requires simulation-specific evidence such as a recreated environment, simulation activity, or role play. The administrative assessment type 'Case studies' is assigned weight 2.0 and remains in the review queue unless stronger case-method evidence appears in the same item.

8. INFS6600 regression checks

Acceptance check	Result
Eight taxonomy categories loaded	PASS
Expected three INFS6600 categories present	PASS
Simulation evaluated independently and not positive	PASS
Administrative case labels remain review-only	PASS
Thresholds labelled provisional	PASS

9. Open configuration points

- Confirm whether the 3.0 positive and 5.0 high-confidence thresholds should remain.
- Confirm the tentative Career readiness weight of 2.0.
- Confirm whether administrative 'Case studies' labels should remain at review weight 2.0.

10. Audit trail

Taxonomy version: 2026-09-01-week4-v2

Official source: <https://www.sydney.edu.au/units/INFS6600/2026-S2C-NE-CC>

Retrieved: 2026-08-31T16:10:06.515477+00:00